

10th Grade Learning Guide: Maximax and Maximin

Learning Criterion

C4: Applies the Maximax and Maximin criteria to make optimistic or conservative decisions without probabilities, selecting the alternative with the highest possible payoff or the best worst-case outcome.

1 Maximax and Maximin

When probabilities are unavailable, decisions can be made by comparing the payoffs for each alternative under each possible state of nature.

	Maximax	Maximin
Focus	Best possible payoff	Worst possible payoff
Attitude	Optimistic	Conservative
First step	Find the maximum in each row	Find the minimum in each row
Final step	Choose the largest maximum	Choose the largest minimum

$$M_i = \max_j P_{ay}(A_i, S_j) \quad A_{\text{Maximax}}^* = \arg \max_i M_i$$

$$m_i = \min_j P_{ay}(A_i, S_j) \quad A_{\text{Maximin}}^* = \arg \max_i m_i$$

2 Solved Example: Coffee Shop

A coffee shop must decide whether to launch a dessert line or keep its current menu. Customer traffic could be high or low.

Alternative	High traffic (S_1)	Low traffic (S_2)
Launch dessert line (A_1)	42	8
Keep current menu (A_2)	28	20

All payoffs are in thousands of dollars.

Step 1: Identify

$A_1 = \text{Launch dessert line}$ $A_2 = \text{Keep current menu}$

$S_1 = \text{High traffic}$ $S_2 = \text{Low traffic}$

Steps 2–3: Complete the analysis

Alternative	S_1	S_2	Max	Min
A_1	42	8	$\max\{42, 8\} = 42$	$\min\{42, 8\} = 8$
A_2	28	20	$\max\{28, 20\} = 28$	$\min\{28, 20\} = 20$

Maximax

Maximin

$42 > 28$

$20 > 8$

$A_{\text{Maximax}}^* = A_1$

$A_{\text{Maximin}}^* = A_2$

Choose the alternative with the highest possible payoff. Choose the alternative with the highest worst-case payoff.

3 Guided Practice: Clothing Store

A clothing store must decide whether to launch a new advertising campaign or keep its current strategy. Sales could be strong or weak.

Alternative	Strong sales (S_1)	Weak sales (S_2)
Launch new campaign (A_1)	55	5
Keep current strategy (A_2)	35	22

All payoffs are in thousands of dollars.

Step 1: Identify

$A_1 =$ _____ $S_1 =$ _____

$A_2 =$ _____ $S_2 =$ _____

Step 2: Complete the table

Alternative	Max	Min
A_1	_____	_____
A_2	_____	_____

Step 3: Decide

Maximax

$$M_1 = \max\{55, 5\} = \underline{\hspace{2cm}}$$

$$M_2 = \max\{35, 22\} = \underline{\hspace{2cm}}$$

$$A_{\text{Maximax}}^* = \underline{\hspace{2cm}}$$

Maximin

$$m_1 = \min\{55, 5\} = \underline{\hspace{2cm}}$$

$$m_2 = \min\{35, 22\} = \underline{\hspace{2cm}}$$

$$A_{\text{Maximin}}^* = \underline{\hspace{2cm}}$$

4 Practice: Restaurant

A restaurant must decide whether to introduce a new lunch menu or keep its current menu. Demand could be high or low.

Alternative	High demand (S_1)	Low demand (S_2)
New lunch menu (A_1)	48	6
Current menu (A_2)	30	21

All payoffs are in thousands of dollars.

Complete the analysis

Alternative	S_1	S_2	Max	Min
A_1	48	6	_____	_____
A_2	30	21	_____	_____

Maximax

$$M_1 = \underline{\hspace{2cm}}$$

$$M_2 = \underline{\hspace{2cm}}$$

$$A_{\text{Maximax}}^* = \underline{\hspace{2cm}}$$

Maximin

$m_2 =$ _____

$m_1 =$ _____

$A^*_{\text{Maximin}} =$ _____

5 Guided Practice: Technology Company

A technology company must choose a strategy for launching a new device. Market demand could be high, medium, or low.

Alternative	High (S_1)	Medium (S_2)	Low (S_3)
Premium version (A_1)	80	35	-20
Standard version (A_2)	60	42	18
Budget version (A_3)	40	32	25
Delay launch (A_4)	15	15	15

All payoffs are in thousands of dollars.

Complete the payoff analysis

Alternative	S_1	S_2	S_3	Max	Min
A_1	80	35	-20	_____	_____
A_2	60	42	18	_____	_____
A_3	40	32	25	_____	_____
A_4	15	15	15	_____	_____

Maximax

$M_1 = \max\{80, 35, -20\} =$ _____

$M_4 = \max\{15, 15, 15\} =$ _____

$M_2 = \max\{60, 42, 18\} =$ _____

$M_3 = \max\{40, 32, 25\} =$ _____

$A^*_{\text{Maximax}} =$ _____

Maximin

$$m_3 = \min\{40, 32, 25\} = \underline{\hspace{2cm}}$$

$$m_1 = \min\{80, 35, -20\} = \underline{\hspace{2cm}}$$

$$m_4 = \min\{15, 15, 15\} = \underline{\hspace{2cm}}$$

$$m_2 = \min\{60, 42, 18\} = \underline{\hspace{2cm}}$$

$$A_{\text{Maximin}}^* = \underline{\hspace{2cm}}$$

6 Independent Practice: Sports Equipment Company

A sports equipment company must choose a production strategy for a new product. Demand could be high, medium, or low.

Alternative	High demand (S_1)	Medium demand (S_2)	Low demand (S_3)
Large production (A_1)	90	30	-15
Medium production (A_2)	65	45	20
Small production (A_3)	45	38	28
Maintain production (A_4)	20	20	20

All payoffs are in thousands of dollars.

Complete the analysis

Alternative	S_1	S_2	S_3	Max	Min
A_1	90	30	-15	<u> </u>	<u> </u>
A_2	65	45	20	<u> </u>	<u> </u>
A_3	45	38	28	<u> </u>	<u> </u>
A_4	20	20	20	<u> </u>	<u> </u>

Maximax

$$M_3 = \underline{\hspace{2cm}}$$

$$M_1 = \underline{\hspace{2cm}}$$

$$M_4 = \underline{\hspace{2cm}}$$

$$M_2 = \underline{\hspace{2cm}}$$

$$A_{\text{Maximax}}^* = \underline{\hspace{2cm}}$$

Maximin

$$m_3 = \underline{\hspace{2cm}}$$

$$m_1 = \underline{\hspace{2cm}}$$

$$m_4 = \underline{\hspace{2cm}}$$

$$m_2 = \underline{\hspace{2cm}}$$

$$A_{\text{Maximin}}^* = \underline{\hspace{2cm}}$$

7 Final Reflection

Maximax asks:

“What is the best that could happen?”

It is appropriate for a decision-maker who is:

Maximin asks:

“What is the worst that could happen?”

It is appropriate for a decision-maker who is:

When the two criteria recommend different alternatives, the decision depends on:
